

SEEWON CHOI

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EDUCATION

University of Pennsylvania

Ph.D. in Computer and Information Science

Advisor: Rajeev Alur

Aug 2023 - Present

Expected May 2028

University of Cambridge

B.A. in Computer Science

Dissertation: *Encoding Representations of Propositional Logic*

Supervisor: Mateja Jamnik

Oct 2019 - Jun 2022

RESEARCH STATEMENT

Despite remarkable advances in AI, concerns about trustworthiness continue to limit its adoption in high-stakes domains such as healthcare. My research addresses these challenges through neurosymbolic AI, combining the strengths of neural learning with symbolic reasoning. I work on scaling neurosymbolic frameworks to complex, real-world applications. My current work focuses on developing foundation models for the early prediction of critical events, including cardiac arrest and sepsis, while explicitly incorporating clinical knowledge into their predictions.

PREPRINTS & PUBLICATIONS

* Equal contribution

CAMEL: An ECG Language Model for Forecasting Cardiac Events 📄 🌐

S Choi*, Mayank Keoliya*, Neelay Velingker*, Alaia Solko-Breslin, Rajeev Alur, Mayur Naik, Eric Wong

Stable Prediction of Adverse Events in Medical Time-Series Data 📄 🌐

S Choi*, Mayank Keoliya*, Rajeev Alur, Mayur Naik, Eric Wong

CTSketch: Compositional Tensor Sketching for Scalable Neurosymbolic Learning 📄 🌐 🔗

S Choi*, Alaia Solko-Breslin*, Rajeev Alur, Eric Wong

NeurIPS 2025

Data-Efficient Learning with Black-box Programs 📄 🌐 🔗

Alaia Solko-Breslin, **S Choi**, Ziyang Li, Neelay Velingker, Rajeev Alur, Mayur Naik, Eric Wong

NeurIPS 2024

Generation of Visual Representations for Multi-Modal Mathematical Knowledge 📄 🌐 🔗

Lianlong Wu*, **S Choi***, Daniel Raggi*, Aaron Stockdill, Grecia Garcia Garcia, Fiorenzo Colarusso, Peter C.H. Cheng, Mateja Jamnik

AAAI 2024 Demo

AWARDS

- Richard Foulke Day Memorial Fellowship (*University of Pennsylvania*, 2023)
- Rosemary Murray Scholarship (*University of Cambridge*, 2019–2021)

TECHNICAL SKILLS & TEACHING

- **Technical Skills:** Python (PyTorch), Rocq, OCaml, Prolog, Haskell, Java
- **Recitation Instructor:** CIS 5200 Machine Learning (Fall 2025)
- **Teaching Assistant:** CIS 4270/5270 Trustworthy Machine Learning (Spring 2026)